

AIRBUS

ABS5377

Issue 3

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November 2023

Airbus Product Standard

Carbon fibre reinforced epoxy prepreg

High performance unidirectional tape 180 °C – curing class

Intermediate modulus fibre

Structural material



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1 Scope

This standard specifies the characteristics of the high performance intermediate modulus carbon fibre reinforced unidirectional tape prepreg, curing 180 °C, for use in aerospace constructions for manufacture of structural component.

2 Normative references

This Airbus Standard incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any these publications apply to this Airbus Standard only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

AIMS05-27-000 Part 1	Technical Specification – High performance carbon fiber reinforced epoxy prepreg – General requirements
AIMS05-27-000 Part 2	Material Specification – High performance carbon fibre reinforced epoxy prepreg – Qualification program / batch release testing – Unidirectional tape / 180 °C curing class – Structural material
AIMS05-27-002	Material Specification – High performance carbon fibre reinforced epoxy prepreg – Unidirectional tape /180 °C-curing class – Intermediate modulus fibre – Structural material

3 Requirements

3.1 Configuration, Dimensions, Tolerances and Mass

The configuration, dimensions, tolerances and mass shall be in accordance with Table 1.

Table 1: Material, tolerances and mass

Code ^{a)}	Material	Thickness per ply (mm) ^{b)}		Resin content (% weight)	Prepreg areal weight (g/m ²)	Fiber areal weight (g/m ²)
		Nominal cpt	Base line cpt			
A	AIMS05-27-002 A	0,129	0,127	34	203	134
B	AIMS05-27-002 B	0,186	0,184	34	294	194
C	AIMS05-27-002 C	0,257	0,254	34	406	268
^{a)} Material code in accordance to AIMS05-27-002 ^{b)} Cured ply thickness may vary depending on specific part geometry and manufacturing process: - Nominal cpt is the result of calculation using nominal values of FAW, and RC for a zero bleed process and a theoretical fiber and resin densities as per corresponding IPS. - Base line cpt is the cpt to be used by design and manufacturing.						

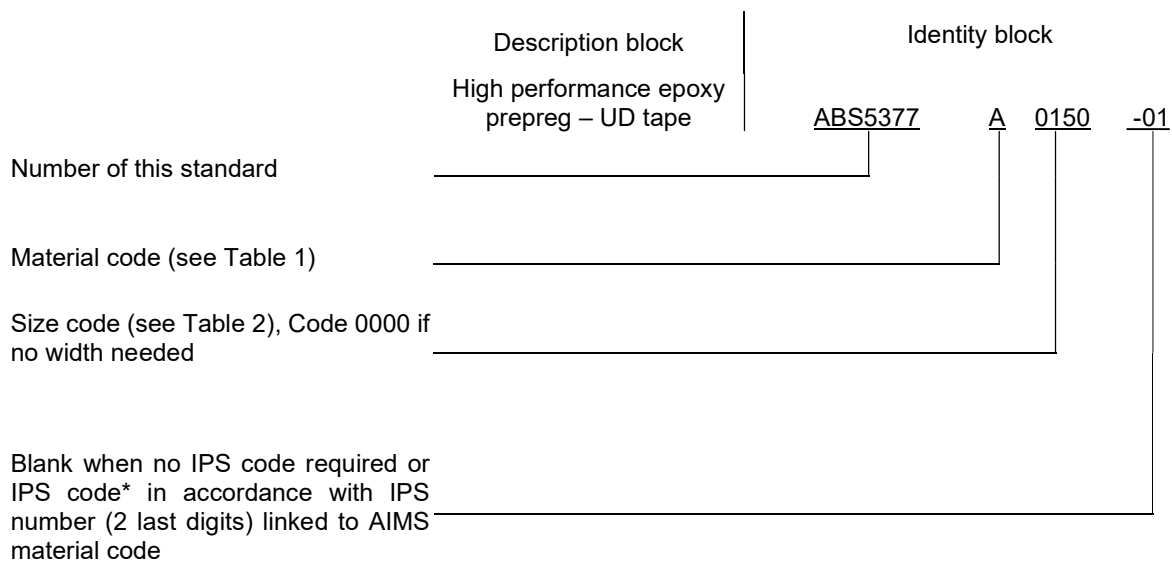
For Dimensions tolerances see Table 2.

Table 2: Dimensions and tolerances

Material	Size code ^{a)}	Width (mm)
Material codes A, B and C	0000 ^{b)}	NA
	0003	3,175 ± 0,127
	0006	6,350 ± 0,127
	0012	12,700 ± 0,127
	0050	50,800 ^{0,000 / -0,400}
	0075	75,00 ^{0,00/-0,75}
	0150	150,00 ^{0,00/-0,75}
	0300	300,00 ^{0,00/-0,75}
	0600	600,00 ± 1,00
NOTE: Length and mass roll could be found in general AIMS or IPS if necessary.		
^{a)} Refer to relevant IPS for applicable size code.		
^{b)} Code 0000 applicable when width requirement is not necessary to be specified.		

4 Designation

This type of standard shall be designated according to the philosophy of the following example:



*Note: Designation with IPS code when required by the design



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5 Marking

See AIMS05-27-000 Part 1.

6 Technical Specification

Technical Specification is AIMS05-27-000 Part 1 and Part 2.

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Record of Revisions

Issue	Clause modified	Description of Modification
1 06.2009		New Standard
2 06.2010	3.1	Table 1 updated.
3 11.2023	All Tables Table 2	Migrated to latest template. Reformatted as required. Footnote numbers changed as letters. Footnote 1) changed as general note "NOTE". Added Size code 0050 New footnote a) added